Anzy Lee

Research Scientist

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EDUCATION

Ph.D in Civil Engineering *2016 - 2020*
Purdue University

MS in Civil and Environmental Engineering *2014 - 2016*
Seoul National University, Republic of Korea

BS in Construction, Urban and Environmental Engineering *2010 - 2014*
Handong Global University, Republic of Korea

RESEARCH/WORK EXPERIENCE

Research Scientist *2020 - 2026*
Adviser: Dr. Belize Lane (Utah State University), Dr. Gregory Pasternack (UC Davis)

- Cooperative Institute for Research to Operations in Hydrology (CIROH) - NOAA *2025 - 2026*
  - Revising spatial and time scales of channel reaches in the hydrofabric to improve OWP flood inundation mapping
- Cooperative Institute for Research to Operations in Hydrology (CIROH) - NOAA *2023 - 2025*
  - Novel Geospatial Architecture of Channel and Floodplain Morphological Attributes within the OWP Hydrofabrics
- California State Water Resources Board *2020 - 2026*
  - Application of methods and models to support the development and implementation of policies for water quality control for cannabis cultivation

Research/Teaching Assistant *2016 - 2020*
Adviser: Dr. Antoine Aubeneau (Purdue University)

- Purdue Research Foundation *2016 - 2020*
  - Riverbed Morphology, Hydrodynamics and Hyporheic Exchange Processes

Research Assistant *2016 - 2020*
Adviser: Dr. Kyung-Duck Suh (Seoul National University)

- National Research Foundation of Korea *2014 - 2015*
  - Design and maintenance of coastal structures considering climate change effects
- Korea Institute of Ocean Science and Technology *2014 - 2015*
  - Development of technology for armor block and crown structure against high waves

SKILLS

Programming & Operating System
- Python, R, Matlab, LaTeX, C++, Linux Bash, Windows Shell, GitHub, Conda, High-Performance Computing (HPC)

Hydraulic & Hydrologic Modeling
- HEC-RAS, TUFLOW, OpenFOAM (RANS, LES), HEC-HMS, MODFLOW, WMS/GMS

Geospatial Analysis
- ArcGIS (ArcPy), QGIS, GDAL, shapely, geopandas, Google Earth

Statistics & Machine Learning
- *Neural networks (PyTorch, TensorFlow), statistical tests, time series analysis, signal processing*

Languages
- *English (Proficient), Korean (Native), Mandarin (Basic)*

PEER-REVIEWED JOURNAL ARTICLES

1. **Lee, A.**, Castejon, J., Patterson, N., Diehl, R., Phillips, C. B., and Lane, B. *(In Preparation)* Probabilistic quantification of within-reach hydraulic geometry variability.
2. Castejon, J., **Lee, A.**, Patterson, N. K., Lane, B., and Phillips, C. B. *(Submitted)* Leveraging High-resolution Topography to Advance Parsimonious Reach-scale Flood Inundation Mapping.
3. **Lee, A.**, White, S., Pasternack, G. B., & Lane, B. (2026). RiverSTICH: Sewing together 3D rivers from only a few loose threads of transect data. *Environmental Modelling & Software*, 106960. doi.org/10.1016/j.envsoft.2026.106960
4. **Lee, A.**, Lane, B., and Pasternack, G. B. (2025). Spectral Slope and Coherence Quantitatively Summarize Nested Topographic Variability Patterns in Rivers. *River Res. Applic.*, 41: 1093-1103. doi:10.1002/rra.4437
5. **Lee, A.**, Lane, B., and Pasternack, G. B. (2023). Identifying key channel variability functions controlling ecohydraulic conditions using synthetic channel archetypes. *Ecohydrology*, e2533. doi:10.1002/eco.2533
6. **Lee, A.**, Cardenas, M. B., Aubeneau, A., and Liu, X. (2022). Hyporheic exchange due to cobbles on sandy beds. *Water Resour. Res.* 58, e2021WR030164. doi:10.1029/2021WR030164
7. **Lee, A.**, Cardenas, M. B., Aubeneau, A., and Liu, X. (2021). Hyporheic Exchange in Sand Dunes Under a Freely Deforming River Water Surface. *Water Resour. Res.* 57, e2020WR028817. doi:10.1029/2020WR028817
8. **Lee, A.**, Cardenas, M. B., and Aubeneau, A. (2020). The Sensitivity of Hyporheic Exchange to Fractal Properties of Riverbeds. *Water Resour. Res.* 56, e2019WR026560. doi:10.1029/2019WR026560
9. Kim, S. W., **Lee, A.**, and Mun, J. (2018). A Surrogate Modeling for Storm Surge Prediction Using an Artificial Neural Network. *J. of Coastal Res.* 84, 866-870. doi:10.2112/SI85-174.1
10. **Lee, A.**, Geem, J. W., and Suh, K. D. (2016). Determination of near-global optimal initial weights of artificial neural network using harmony search algorithm: Application to breakwater armor stones. *Appl. Sci.* 6(6), 164. doi:10.3390/app6060164
11. **Lee, A.**, Kim, S. E., and Suh, K. D. (2016). An easy way to use artificial neural network model for calculating stability number of rock armor. *Ocean Eng.* 127, 349-356. doi:10.1016/j.oceaneng.2016.10.013

AWARD & SERVICE

Peer Reviewer — 2019 - Current
River Res. Applic., Earth Surf. Process. Landf., Water Resour. Res., J. Hydrol., J. Hydraul. Eng.
Dorothy Faye Dunn Fellowship, *Purdue University* — 2019
Delleur Award, *Purdue University* — 2017, 2018
Merit Scholarship (Top 5%), *Handong University* — 2012

TALKS & CONFERENCE PROCEEDINGS

1. Villalobos, J.C., **Lee, A.**, Patterson, N., Stieve, J., Lane, B., and Phillips, C.B. (2026). A terrain-based metric for predicting HAND Flood Inundation Mapping Skill. *CIROH Training and Developers Conference*. May 2026, Salt Lake City, United States.
2. Phillips, C.B., Villalobos, J.C., Patterson, N., **Lee, A.**, Bower, J., Diehl, R.M., Masteller, C.C. & Lane, B., (2025). Leveraging River Channel Geometry to Enable Probabilistic Flood Inundation Forecasts Across Scales. *AGU 2025 Fall Meeting*. Dec 2025, New Orleans, United States.

3 **Lee, A.**, Villalobos, J. C., Patterson, N., Diehl, R. M., Phillips, C. B., & Lane, B. A. A. (2025). Probabilistic quantification of within-reach hydraulic geometry. *AGU 2025 Fall Meeting*. Dec 2025, New Orleans, United States.

4 Stieve, J., Lane, B., and **Lee, A.** *(Submitted to ISE 2026).* Generating Syntetic Terrain Models from Sparse Historic Datasets for Ecohydraulic Assessment.

5 **Lee, A.**, Castejon, J., Patterson, N., Diehl, R., Phillips, C. B., and Lane, B. *(Accepted to AGU 2025. Oral Presentation).* Probabilistic quantification of within-reach hydraulic geometry.

6 Castejon Villalobos, J., **Lee, A.,**, Lane, B., and Phillips, C. B. (2024). Accurate River Channel Representation Within a HAND-y Method for Flood Inundation Mapping. *AGU 2024 Fall Meeting.* Dec 2024, Washington D.C., United States.

7 Patterson, N., Castejon Villalobos, J., **Lee, A.**, Lane, B., Diehl, R. M., and Phillips, C. B. (2024). Leveraging High-Resolution Topography to Determine the Bankfull Channel. *AGU 2024 Fall Meeting.* Dec 2024, Washington D.C., United States.

8 **Lee, A.**, Lane, B., and Pasternack, G. B. (2022). Developing Archetypal River Corridor Terrain Models for Various Channel Types. *AGU 2022 Fall Meeting.* Dec 2022, Chicago, United States.

9 **Lee, A.**, Lane, B., Pasternack, G. B., and Sandoval-Solis, S. (2021). Identifying key geomorphic parameters characterizing eco-hydraulic responses of river channels using RiverBuilder. *AGU 2021 Fall Meeting.* Dec 2021, New Orleans, United States.

10 **Lee, A.**, Cardenas, M. B., and Aubeneau, A. (2018). Investigation of hyporheic exchange in channels with high Froude Number flows: the importance of free surface water elevation changes *AGU 2018 Fall Meeting.* Dec 2018, Washington, D.C., United States.

11 Aubeneau, A. and **Lee, A.** (2018). Aris method for (reactive) transient storage models. *AGU 2018 Fall Meeting.* Dec 2018, Washington, D.C., United States.

12 **Lee, A.** and Aubeneau, A. (2017). 3D Numerical Modeling of Hyporheic Exchange Processes in Fractal Riverbed. *AGU 2017 Fall Meeting.* Dec 2017, New Orleans, United States.

DIGITAL PRODUCTS

Lee, A. and Lane, B. (2024). HAND-FIM Assessment Tools [Python]. GitHub. Link.
Lee, A. (2024). Width Extraction Tools [Python]. GitHub. Link.
Lee, A. (2024). Habitat Analysis Tools [Python]. GitHub. Link.
Lee, A. (2024). Spectral Analysis Tools for Geomorphic Variability Functions [MATLAB]. GitHub. Link.
Lee, A. (2020). hyporheicScalarInterFoam [C++, OpenFOAM]. HydroShare. Link.

TEACHING & MENTORING

Research Mentoring, *Utah State University* *2020 - 2026*
Undergraduate- Maddie Witte, *Graduate*- Steve White, Jared Stieve
Lab Instructor for Elementary Fluid Mechanics, *Purdue University* *2019*